

A Permutation-Null Framework for Conservative Feature Selection in Redundant Learning Problems

Elliot M. de Florinier

Email: elliot@deflorinier.me.uk

Abstract

Feature selection is widely employed to improve model generalisation and interpretability in high-dimensional learning problems. However, many existing approaches rely on deterministic feature importance rankings that lack statistical interpretation and frequently lead to over-pruning, particularly in engineering and scientific datasets characterised by feature correlation, redundancy, and limited sample sizes.

This paper proposes a permutation-null feature selection framework that evaluates feature relevance by comparing observed model-derived importance values against empirical null distributions obtained under feature–label decoupling. Feature selection is formulated as a hypothesis testing problem, retaining predictors whose importance exceeds noise-level expectations. To prevent excessive dimensionality reduction, the framework incorporates an adaptive threshold selection procedure and a conservative prune-ratio constraint.

Experimental evaluation across datasets with varying redundancy characteristics demonstrates that the proposed approach preserves predictive performance on well-conditioned datasets while improving stability and generalisation in the presence of distractor features. Results indicate that feature selection does not universally improve accuracy; however, statistically grounded noise-rejection strategies offer measurable benefits in redundancy-dominated regimes.

Index Terms

Feature selection, permutation testing, null importance, dimensionality reduction, model stability, redundancy.

1. Introduction

High-dimensional datasets are increasingly common across engineering and scientific disciplines, including materials informatics, sensor-based monitoring, and manufacturing process modelling. Such datasets often exhibit strong feature correlation, redundant descriptors, and comparatively small sample sizes, posing significant challenges for reliable model generalisation.

Feature selection (FS) is frequently applied to address these challenges by reducing dimensionality, improving interpretability, and mitigating overfitting. Conventional FS methods are typically classified as filter, wrapper, or embedded approaches [1]. Despite their widespread use, many techniques rely on deterministic feature importance rankings that lack explicit statistical interpretation. Consequently, features exhibiting non-zero importance due to random variation may be retained, while weak but informative predictors may be discarded.

Empirical studies have shown that feature selection does not guarantee performance improvement and may degrade predictive accuracy when applied indiscriminately [2]. These observations motivate the need for feature selection strategies that are conservative, interpretable, and capable of distinguishing genuine signal from noise.

This paper introduces a permutation-null feature selection framework that explicitly models the distribution of feature importance expected under random association with the target variable. Rather than ranking features absolutely, relevance is assessed through deviation from an empirically derived null distribution. The proposed method is designed not as an accuracy maximisation strategy, but as a statistically grounded noise-rejection mechanism that adapts its behaviour to dataset structure.

2. Related Work

2.1 Filter Methods

Filter methods evaluate features independently of the learning algorithm, using criteria such as correlation, mutual information, or distance-based relevance measures [3], [4]. While computationally efficient, these approaches are inherently univariate and fail to capture feature interactions or conditional dependencies.

2.2 Wrapper Methods

Wrapper methods, including recursive feature elimination (RFE), assess subsets of features through repeated model training and evaluation [5]. Although capable of capturing interactions, wrapper approaches are computationally expensive and often exhibit high variance, particularly in small-sample regimes.

2.3 Embedded Methods

Embedded techniques integrate feature selection into model optimisation, such as L1-regularised regression [6] and tree-based impurity importance measures [7]. While attractive in practice, impurity-based importance scores are known to be biased toward continuous or high-cardinality features and may distribute importance arbitrarily among correlated predictors [8].

A common limitation across these approaches is the absence of explicit modelling of importance arising from noise, motivating statistically principled alternatives.

3. Methodology

3.1 Problem Formulation

Let $X \in \mathbb{R}^{n \times d}$ denote a dataset with n samples and d features, and let $y \in \mathbb{R}^n$ represent the target variable. For each feature x_j , a supervised learning model produces an observed importance score I_j^{real} .

Rather than interpreting this value directly, feature relevance is formulated as a hypothesis testing problem:

$$H_0: I_j^{real} \sim I_j^{null}$$

where I_j^{null} denotes the distribution of importance values expected under random association between features and targets.

3.2 Null Importance Generation

Empirical null distributions are generated through feature–label decoupling. For each iteration, feature columns are independently permuted, destroying predictive relationships while preserving marginal distributions. A Random Forest classifier is trained on the permuted dataset, and feature importances are recorded. Repeating this process yields a null distribution for each feature:

$$\{I_j^{(k)}\}_{k=1}^N$$

These distributions characterise importance attributable solely to noise.

3.3 Statistical Decision Rule

For each feature, the null distribution mean μ_j and standard deviation σ_j are estimated. Feature relevance is determined using the criterion:

$$I_j^{real} > \mu_j + k\sigma_j$$

where k controls pruning aggressiveness. Larger values of k correspond to increasingly conservative selection.

3.4 Adaptive Threshold Selection

To avoid reliance on a fixed threshold, candidate values $k \in \{0,1,2,3\}$ are evaluated using an internal validation split when no threshold is specified. The value yielding the highest validation accuracy is selected automatically.

3.5 Conservative Pruning Constraint

To prevent excessive dimensionality reduction, a prune-ratio constraint is imposed, ensuring that a minimum fraction of features is retained. If thresholding selects fewer features than permitted, additional predictors are retained according to descending observed importance.

4. Experimental Setup

4.1 Datasets

Datasets with varying redundancy characteristics were selected to evaluate method behaviour under diverse conditions.

Table 1. Summary of datasets used in experimental evaluation

<i>Dataset</i>	<i>Features</i>	<i>Classes</i>	<i>Evaluations</i>
<i>Breast Cancer (low redundancy)</i>	30	2	275
<i>Digits (moderate redundancy)</i>	64	10	275
<i>HAR (moderate–high redundancy)</i>	561	6	25
<i>Madelon (heavy redundancy)</i>	500	2	275

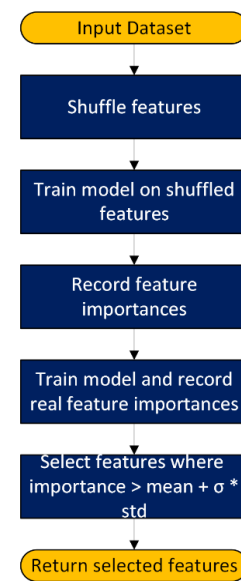


Figure 1. Overview of the proposed permutation-null feature selection framework

4.2 Baseline Methods

The proposed approach was compared against:

- No feature selection (baseline)
- Mutual Information (filter method)
- Recursive Feature Elimination (wrapper method)
- Embedded selection using tree-based importance thresholds

4.3 Evaluation Protocol

Repeated train–test splits with varying random seeds were employed. Feature selection was performed exclusively on training data to avoid information leakage. Performance was assessed using test accuracy, paired accuracy improvement relative to baseline, and variance across runs.

5. Results

5.1 Feature Retention Behaviour

Table 2. Average retained dimensionality (mean $\pm$ standard deviation)

<i>Dataset</i>	<i>Method</i>	<i>Retained Features</i>
Breast Cancer (low redundancy)	<i>Baseline</i>	<i>30.0 $\pm$ 0.0</i>
	<i>ElliotFS</i>	9.2 $\pm$ 1.1
	<i>L1</i>	<i>30.0 $\pm$ 0.0</i>
	<i>Mutual Information</i>	<i>30.0 $\pm$ 0.0</i>
	<i>RFE</i>	<i>30.0 $\pm$ 0.0</i>
Digits (moderate redundancy)	<i>Baseline</i>	<i>64.0 $\pm$ 0.0</i>
	<i>ElliotFS</i>	19.9 $\pm$ 1.5
	<i>L1</i>	<i>32.0 $\pm$ 0.0</i>
	<i>Mutual Information</i>	<i>32.0 $\pm$ 0.0</i>
	<i>RFE</i>	<i>32.0 $\pm$ 0.0</i>

HAR (moderate–high redundancy)	<i>Baseline</i>	561.0 ± 0.0
	<i>ElliotFS</i>	123.6 ± 3.3
	<i>L1</i>	50.0 ± 0.0
	<i>Mutual Information</i>	50.0 ± 0.0
	<i>RFE</i>	50.0 ± 0.0
Madelon (heavy redundancy)	<i>Baseline</i>	500.0 ± 0.0
	<i>ElliotFS</i>	100.0 ± 0.0
	<i>L1</i>	50.0 ± 0.0
	<i>Mutual Information</i>	50.0 ± 0.0
	<i>RFE</i>	50.0 ± 0.0

5.2 Predictive Performance

Table 3. Test accuracy (mean ± standard deviation)

<i>Dataset</i>	<i>Method</i>	<i>Test Accuracy</i>
<i>Breast Cancer (low redundancy)</i>	<i>Baseline</i>	0.958 ± 0.017
	<i>ElliotFS</i>	0.949 ± 0.016
	<i>L1</i>	0.944 ± 0.017
	<i>Mutual Information</i>	0.958 ± 0.017
	<i>RFE</i>	0.958 ± 0.017
<i>Digits (moderate redundancy)</i>	<i>Baseline</i>	0.977 ± 0.005
	<i>ElliotFS</i>	0.960 ± 0.008
	<i>L1</i>	0.975 ± 0.006
	<i>Mutual Information</i>	0.974 ± 0.006
	<i>RFE</i>	0.975 ± 0.005
<i>HAR (moderate–high redundancy)</i>	<i>Baseline</i>	0.978 ± 0.001
	<i>ElliotFS</i>	0.977 ± 0.002

Madelon (heavy redundancy)

L1	0.977 ± 0.001
Mutual Information	0.938 ± 0.004
RFE	0.976 ± 0.004
Baseline	0.714 ± 0.019
ElliotFS	0.832 ± 0.013
L1	0.806 ± 0.016
Mutual Information	0.716 ± 0.032
RFE	0.859 ± 0.013

5.3 Accuracy Improvement Relative to Baseline

Figure 2. Mean accuracy improvement relative to baseline

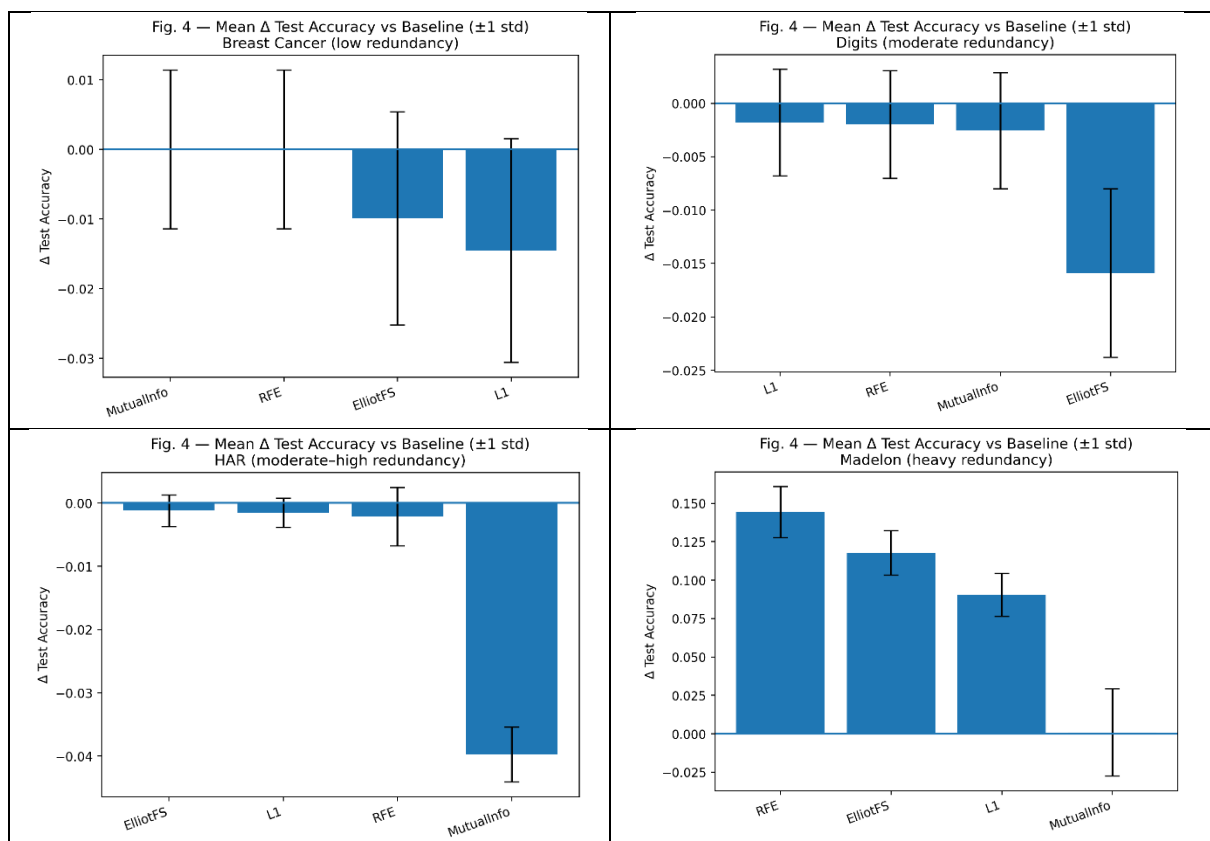


Table 4. Paired accuracy improvement vs baseline

Dataset	Method	Δ Accuracy
---------	--------	-------------------

Breast Cancer (low redundancy)	<i>Mutual Information</i>	0.000 ± 0.011
	<i>RFE</i>	0.000 ± 0.011
	<i>ElliotFS</i>	-0.010 ± 0.015
	<i>L1</i>	-0.015 ± 0.016
Digits (moderate redundancy)	<i>L1</i>	-0.002 ± 0.005
	<i>RFE</i>	-0.002 ± 0.005
	<i>Mutual Information</i>	-0.003 ± 0.005
	<i>ElliotFS</i>	-0.016 ± 0.008
HAR (moderate–high redundancy)	<i>ElliotFS</i>	-0.001 ± 0.003
	<i>L1</i>	-0.002 ± 0.002
	<i>RFE</i>	-0.002 ± 0.005
	<i>Mutual Information</i>	-0.040 ± 0.004
Madelon (heavy redundancy)	<i>RFE</i>	$+0.144 \pm 0.017$
	<i>ElliotFS</i>	$+0.118 \pm 0.014$
	<i>L1</i>	$+0.090 \pm 0.014$
	<i>Mutual Information</i>	$+0.001 \pm 0.028$

6. Discussion

The experimental results indicate that feature selection does not universally improve predictive accuracy. On well-conditioned datasets exhibiting limited redundancy, dimensionality reduction provided negligible benefit and occasionally induced minor performance degradation. This behaviour is consistent with theoretical expectations and highlights the risks of indiscriminate pruning.

In contrast, datasets containing correlated or distractor features exhibited measurable improvements under the proposed permutation-null framework. Notably, the method demonstrated reduced variance across repeated train–test splits compared to competing approaches, indicating improved stability under sampling uncertainty.

These findings support the interpretation of the proposed method as a conservative noise-rejection mechanism rather than an accuracy maximisation strategy.

7. Limitations

The proposed framework relies on Random Forest impurity importance, which is subject to known biases under correlated features. Computational cost is higher than that of simple filter methods due to repeated null generation. Additionally, the approach does not guarantee optimal feature subsets under all conditions.

Future work may explore permutation importance measures, false discovery rate control, and correlation-aware group selection.

8. Conclusion

This paper presented a permutation-null feature selection framework that evaluates feature relevance through deviation from empirically derived noise distributions. By incorporating adaptive threshold selection and conservative pruning constraints, the method demonstrates stable and interpretable behaviour across datasets with varying redundancy levels.

Experimental results show that while feature selection does not universally improve accuracy, statistically grounded noise-rejection strategies can enhance generalisation stability in redundancy-dominated learning problems. The proposed approach is particularly suited to engineering and scientific domains where robustness and interpretability are prioritised.

References

- [1] I. Guyon and A. Elisseeff, “An Introduction to Variable and Feature Selection,” *J. Mach. Learn. Res.*, vol. 3, pp. 1157–1182, 2003.
- [2] R. Kohavi and G. H. John, “Wrappers for Feature Subset Selection,” *Artif. Intell.*, vol. 97, no. 1–2, pp. 273–324, 1997.
- [3] H. Peng, F. Long, and C. Ding, “Feature Selection Based on Mutual Information Criteria,” *IEEE TPAMI*, vol. 27, no. 8, pp. 1226–1238, 2005.
- [4] K. Kira and L. A. Rendell, “A Practical Approach to Feature Selection,” in *Proc. ICML*, 1992.
- [5] I. Guyon et al., “Gene Selection Using SVM,” *Mach. Learn.*, vol. 46, pp. 389–422, 2002.
- [6] R. Tibshirani, “Regression Shrinkage and Selection via the Lasso,” *JRSS-B*, vol. 58, no. 1, pp. 267–288, 1996.
- [7] L. Breiman, “Random Forests,” *Mach. Learn.*, vol. 45, pp. 5–32, 2001.
- [8] G. Louppe et al., “Understanding Variable Importances in Forests of Randomized Trees,” in *NeurIPS*, 2013.